

Appendix

0.1 Notation and conventions

N_x, N_y	number of tones per axis, indices i, j
W	frequency span per axis, tone spacing $\Delta f_x = W/(N_x - 1)$
$\phi_{x,i}, \phi_{y,j}$	tone phases, free parameters at the AWG
$a_{x,i}, a_{y,j}$	tone <i>power</i> weights, $A_s = \sqrt{a_{x,i} a_{y,j}}$
$u(r)$	focal mode, normalised to $u(0) = 1$
f_0, T_0	fundamental beat frequency and period
k_s	frequency order of spot s , H_m group field of order m
D_d	Fourier coefficient of the intensity, beat order d
C	crest factor of the RF signal per axis
η, α, P_s	diffraction efficiency, peak efficiency, saturation power

The amplitude ratios r_x, r_y are *power* ratios. The RF voltage amplitudes therefore carry the square root,

$$a = [\varrho, 1, \dots, 1, \varrho] \text{ in voltage, } \quad [r, 1, \dots, 1, r] \text{ in power, } \quad \varrho := \sqrt{r}. \quad (1)$$

0.2 Beating model

$$f_s = f_{x,i} + f_{y,j} = f_{\min} + k_s f_0, \quad f_0 = \frac{W}{\text{lcm}(N_x - 1, N_y - 1)}, \quad T_0 = \frac{1}{f_0}. \quad (2)$$

$$g_s(\mathbf{r}) = A_s u(\mathbf{r} - \mathbf{c}_s), \quad \phi_s = \phi_{x,i} + \phi_{y,j}, \quad H_m(\mathbf{r}) = \sum_{s: k_s=m} g_s(\mathbf{r}) e^{i\phi_s}. \quad (3)$$

Only $N_x + N_y$ of the $N_x N_y$ spot phases are free. Dropping the global factor $e^{i2\pi f_{\min} t}$, which cancels in $|E|^2$,

$$\boxed{E(\mathbf{r}, t) = \sum_m H_m(\mathbf{r}) e^{i2\pi m f_0 t}, \quad I(\mathbf{r}, t) = \sum_d D_d(\mathbf{r}) e^{i2\pi d f_0 t}} \quad (4)$$

$$D_d(\mathbf{r}) = \sum_{k_s - k_{s'} = d} g_s(\mathbf{r}) g_{s'}(\mathbf{r}) e^{i(\phi_s - \phi_{s'})}, \quad D_{-d} = D_d^*, \quad f_{\text{beat}} = d f_0. \quad (5)$$

Time average over one period T_0 :

$$\langle I \rangle_t = D_0 = \sum_s g_s^2 + 2 \sum_{\substack{s < s' \\ k_s = k_{s'}}} g_s g_{s'} \cos(\phi_s - \phi_{s'}), \quad (6)$$

so the time average equals the incoherent sum only if every group contains a single spot; a degenerate pair contributes a static term that vanishes for $\phi_s - \phi_{s'} = \pm\pi/2$. Different orders oscillate at different frequencies and are orthogonal over T_0 , so the variances add:

$$\text{Var}_t[I] = \sum_{d \neq 0} |D_d|^2 = \sum_{d > 0} \sigma_d^2, \quad \sigma_d^2 := 2 |D_d|^2, \quad (7)$$

$$U_t(\mathbf{r}) = \frac{\sigma_I(\mathbf{r})}{\langle I \rangle_t(\mathbf{r})}, \quad \bar{U}_t = \langle U_t^2 \rangle_R^{1/2} = \left(\sum_{d > 0} \bar{u}_d^2 \right)^{1/2}, \quad \bar{u}_d = \left\langle \frac{\sigma_d^2}{\langle I \rangle_t^2} \right\rangle_R^{1/2}, \quad (8)$$

with $\langle \cdot \rangle_R$ the average over the spot square R , the same region as for U_h . The extreme order $d = k_{\max} - k_{\min}$ is produced by a single pair of corner spots, so $|D_{d_{\max}}|$ cannot be changed by any choice of tone phases.

0.3 Choice of the tone phases

0.3.1 The centring criterion

The atom must see a pattern that is symmetric at *every* instant, not merely on average:

$$I(-\mathbf{r}, t) = I(\mathbf{r}, t) \quad \forall t \quad \Longleftrightarrow \quad E(-\mathbf{r}, t) = e^{i\alpha(t)} E^*(\mathbf{r}, t), \quad (9)$$

with one and the same $\alpha(t)$ for the whole field. Starting from

$$E(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{-i2\pi(f_n + g_m)t} e^{i(\varphi_n + \psi_m)}, \quad (10)$$

three symmetries of the profile enter, with the mirror index $\bar{n} = N - 1 - n$:

$$\mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{c}_{nm} \quad (\text{centred spot grid}), \quad (11)$$

$$a_{\bar{n}} = a_n, \quad b_{\bar{m}} = b_m \quad (\text{amplitudes } [\varrho, 1, \dots, 1, \varrho]), \quad (12)$$

$$f_{\bar{n}} = F - f_n, \quad g_{\bar{m}} = G - g_m, \quad F = f_n + f_{\bar{n}} = 2f_{\text{off}} + (N - 1)\Delta f. \quad (13)$$

Evaluating E at $-\mathbf{r}$ and relabelling the sum by the mirror index:

Step 1 (positions, u even): $u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}}) = u(-(\mathbf{r} - \mathbf{c}_{nm})) = u(\mathbf{r} - \mathbf{c}_{nm})$, the envelope factor is reproduced.

Step 2 (amplitudes): $a_{\bar{n}} b_{\bar{m}} = a_n b_m$, unchanged.

Step 3 (frequencies), the decisive one:

$$e^{-i2\pi(f_{\bar{n}} + g_{\bar{m}})t} = e^{-i2\pi(F + G)t} \cdot e^{+i2\pi(f_n + g_m)t}, \quad (14)$$

i.e. the frequency mirror does not reproduce the time factor, it *conjugates* it, up to one global factor common to all spots.

Step 4 (collect and compare):

$$E(-\mathbf{r}, t) = e^{-i2\pi(F + G)t} \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n + g_m)t} e^{i(\varphi_{\bar{n}} + \psi_{\bar{m}})}, \quad (15)$$

$$E^*(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n + g_m)t} e^{-i(\varphi_n + \psi_m)}. \quad (16)$$

Both sums run over the same basis functions with the same real coefficients and differ only by a constant phase per term. They are proportional, and hence Eq. (9) holds, if and only if $e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})} = e^{i\Theta} e^{-i(\varphi_n+\psi_m)}$ for all n, m with a *single* constant Θ . Separating the two axes,

$$\boxed{\varphi_n + \varphi_{N_x-1-n} = c_x \quad \forall n, \quad \psi_m + \psi_{N_y-1-m} = c_y \quad \forall m} \quad (17)$$

Mirror-symmetric phases, $\varphi_{\bar{n}} = \varphi_n$, are a different condition and do not centre the pattern; they leave residual centroid offsets of 44 nm to 84 nm against $\sigma = 108$ nm, whereas Eq. (17) gives 0.00 nm.

0.3.2 Structure of the admissible family

With $\varphi_n = c/2 + \chi_n$, Eq. (17) becomes

$$\chi_{\bar{n}} = -\chi_n \quad \implies \quad \dim(\text{family}) = \left\lfloor \frac{N}{2} \right\rfloor \quad \text{instead of} \quad N-1. \quad (18)$$

For odd N the map $n \mapsto \bar{n}$ has a fixed point, so the middle tone is paired with *itself* and its phase is pinned:

$$2\varphi_{(N-1)/2} = c \quad \implies \quad \varphi_{(N-1)/2} = \frac{c}{2} \quad \text{or} \quad \frac{c}{2} + \pi. \quad (19)$$

A ramp $\chi_n = \delta \hat{n}$ with $\hat{n} := n - (N-1)/2$ satisfies Eq. (18) automatically and is a pure time shift of the envelope,

$$A_\delta(t) = A_0(t + \tau), \quad \tau = \frac{\delta}{2\pi \Delta f}, \quad (20)$$

hence crest neutral and redundant with the trigger t_0 . One of the $\lfloor N/2 \rfloor$ dimensions is therefore always inactive:

$$\dim(\text{crest-relevant}) = \left\lfloor \frac{N}{2} \right\rfloor - 1 \xrightarrow{N=3} 0. \quad (21)$$

0.3.3 Crest factor

$$C = \frac{\max_t |s(t)|}{\text{rms}_t[s(t)]} = \sqrt{2} \frac{\max_t |A(t)|}{\sqrt{\sum_n a_n^2}}, \quad \sqrt{2} \leq C \leq \sqrt{2} \frac{\sum_n a_n}{\sqrt{\sum_n a_n^2}} \xrightarrow{a_n \equiv 1} \sqrt{2N}, \quad (22)$$

both operations taken over a full envelope period; the denominator is phase independent [1, 2]. For $N_x = 3$, inserting Eq. (17) as $\varphi = (\frac{c}{2} - \chi, \frac{c}{2} \text{ or } \frac{c}{2} + \pi, \frac{c}{2} + \chi)$ and factoring out $e^{ic/2}$, with $\theta := 2\pi \Delta f_x t$,

$$\boxed{A(t) e^{-ic/2} = \varrho_x e^{-i(\chi+\theta)} \pm 1 + \varrho_x e^{+i(\chi+\theta)} = 2\varrho_x \cos(\theta + \chi) \pm 1} \quad (23)$$

The only free parameter appears *inside* the cosine, and $\cos$ sweeps $[-1, 1]$ over one period regardless of χ , so

$$\max_t |A| = \max_{v \in [-1, 1]} |2\varrho_x v \pm 1| = 2\varrho_x + 1 = \sum_n a_n \quad \text{for all } c, \chi \text{ and both branches}, \quad (24)$$

which is the upper bound of Eq. (22). With $r_x = 0.97$ and $\varrho_x = \sqrt{r_x} = 0.9848858$,

$$C_x = \sqrt{2} \frac{2\varrho_x + 1}{\sqrt{2r_x + 1}} = \sqrt{2} \frac{2.9697716}{1.7146428} = 2.4494 \quad (\sqrt{2N} = \sqrt{6} = 2.4495). \quad (25)$$

For $N_y = 4$ there is no fixed point and Eq. (17) leaves $\psi = [0, b, c_y - b, c_y]$ with two parameters, one of them crest relevant. At $c_y = 114^\circ$:

b	ψ [deg]	$\max A $	C_y
38°	[0, 38, 76, 114]	4.154	2.8265 (ramp, $= \sum_n a_n$)
60°	[0, 60, 54, 114]	4.022	2.7364
98°	[0, 98, 16, 114]	3.228	2.1963
114°	[0, 114, 0, 114]	3.639	2.4761

A power amplifier clips the instantaneous voltage and reaches compression long before any thermal limit [3, 4], so with $P_{\text{peak}} = C^2 P_{\text{avg}}$ and $P_{\text{peak}} \leq P_{\text{1dB}}$,

$$\boxed{P_{\text{avg}} \leq \frac{P_{\text{1dB}}}{C^2}} \quad P_{\text{1dB}} = 36 \text{ dBm} = 3.98 \text{ W [5]}. \quad (26)$$

	C	C^2	$P_{\text{avg}} \leq P_{\text{1dB}}/C^2$
x axis, $N_x = 3$, forced by Eq. (21)	2.4494	6.00	0.66 W
y axis, $N_y = 4$, $b = 98^\circ$, minimum in the family	2.1963	4.82	0.83 W

0.4 Pulse area for a given profile power

The optical power P in the profile fixes the scale s in $I = s\tilde{I}$, where $\tilde{I} = |E|^2$ is the simulated intensity in units of $u(0) = 1$:

$$P = \int \langle I \rangle_t d^2r = s \tilde{P}_{\text{prof}}, \quad \tilde{P}_{\text{prof}} = \sum_s a_s P_1 + 2 \sum_{\substack{s < s' \\ k_s = k_{s'}}} A_s A_{s'} \cos(\phi_s - \phi_{s'}) O(\varrho_{ss'}), \quad (27)$$

with the single-spot power $P_1 = \int u^2 d^2r$ and the mode autocorrelation O evaluated at the spot separation $\varrho_{ss'}$ [6–8]:

$$P_1^G = \frac{\pi w_0^2}{2}, \quad O^G(\varrho) = P_1^G e^{-\varrho^2/2w_0^2}, \quad P_1^A = \frac{4\pi}{k^2}, \quad O^A(\varrho) = P_1^A u(\varrho), \quad (28)$$

$k = 3.8317/(1.4830 w_0)$. The Airy autocorrelation is the mode itself, sign included. Adiabatic elimination of the $5P$ manifold, with signed dipole matrix elements and hyperfine shifts from ARC [9, 10] and the far detuned scaling of [11], makes all three atomic quantities linear in intensity:

$$\Omega = C_{\text{rabi}} I, \quad \delta_{\text{LS}} = C_{\text{shift}} I, \quad \Gamma_{\text{sc}} = C_{\text{scatter}} I, \quad \varepsilon_{\text{LS}} = \frac{\delta_{\text{LS}}}{\Omega} = \text{const.} \quad (29)$$

The area of a rectangular pulse on $[t_0, t_0 + T_p]$ is therefore

$$\boxed{\theta(\mathbf{r}, t_0) = C_{\text{rabi}} P \frac{\int_{t_0}^{t_0+T_p} \tilde{I}(\mathbf{r}, t) dt}{\int \langle \tilde{I} \rangle_t d^2r}} \quad (30)$$

in which the arbitrary mode normalisation cancels. Inserting Eq. (4) and integrating each order over the window,

$$\int_{t_0}^{t_0+T_p} e^{i2\pi df_0 t} dt = T_p \text{sinc}(df_0 T_p) e^{i2\pi df_0 (t_0+T_p/2)}, \quad (31)$$

$$\theta(\mathbf{r}, t_0) = \frac{C_{\text{rabi}} P T_p}{\bar{P}_{\text{prof}}} \left[D_0 + 2 \sum_{d>0} |D_d| \operatorname{sinc}(d f_0 T_p) \cos\left(2\pi d f_0 \left(t_0 + \frac{T_p}{2}\right) + \arg D_d\right) \right]. \quad (32)$$

For $T_p \gg T_0$ every $d \neq 0$ is damped and the pulse sees the time average; for $T_p \ll T_0$ the sinc is unity and the pulse samples the beating, so θ depends on the trigger t_0 . Averaging over the atom, with $W(\mathbf{r})$ the thermal position distribution of width $\sigma^2 = \frac{\hbar}{2m\omega_r} \coth \frac{\hbar\omega_r}{2k_B T}$ [12],

$$c_d = \int W D_d d^2r, \quad U(\theta) = \frac{\sigma_W(\theta)}{\langle \theta \rangle_W}, \quad \langle p_{\uparrow} \rangle_W = \int W \sin^2 \frac{\theta}{2} d^2r \neq \sin^2 \frac{\langle \theta \rangle_W}{2}. \quad (33)$$

Since θ is linear in the D_d , a full scan over t_0 is a single matrix-vector product in the c_d .

$\bar{P}_{\text{prof}}$	$2.615 \times 10^{-11} \text{ m}^2$	$\langle \theta \rangle_W$ at best trigger	11.54π
C_{rabi}	$11.66 \frac{\text{rad}}{\text{s}} / (\text{W}/\text{m}^2)$	$U(\theta)$ there	0.72 %
ε_{LS}	-0.061	P for a 1 μs π pulse	0.87 μW

Working point: 3×4 , Airy mode, $w_0 = 1.04 \mu\text{m}$, $r_x/r_y = 0.97/1.16$, $\Delta = +50 \text{ GHz}$, $P = 10 \mu\text{W}$ in the profile, $T_p = 1 \mu\text{s}$, $\sigma = 108 \text{ nm}$.

0.5 Diffraction efficiency and profile power

Single grating in the Bragg regime, with $\Delta n \propto \sqrt{P_{\text{ac}}}$ from the photoelastic effect [13–15]:

$$\eta(P) = \alpha(\nu) \sin^2 \left(\frac{\pi}{2} \sqrt{\frac{P}{P_s(\nu)}} \right), \quad \kappa = \frac{\pi}{\lambda} \sqrt{\frac{M_2 L}{2H}} \sqrt{\zeta}, \quad P_s = \left(\frac{\pi}{2\kappa} \right)^2. \quad (34)$$

With N tones the gratings share one undiffracted wave; the coupled wave equations [16, 17] reduce exactly to a two wave problem and give [18, 19]

$$\boxed{\eta_{\text{tot}} = \alpha \sin^2 \left(\frac{\pi}{2} \sqrt{\frac{P_{\text{tot}}}{P_s}} \right), \quad \eta_n = \frac{P_n}{P_{\text{tot}}} \eta_{\text{tot}}, \quad P_{\text{tot}} = \sum_n P_n} \quad (35)$$

so the total efficiency depends only on the total RF power and not on its distribution among the tones, and the tone phases do not enter at all. Combining with Eq. (26),

$$\boxed{\eta_{\text{axis}}(C) = \alpha \sin^2 \left(\frac{\pi}{2C} \sqrt{\frac{P_{\text{1dB}}}{P_s}} \right), \quad \eta_{\text{profile}} = \eta_x \eta_y \Xi} \quad (36)$$

with the measured cross axis term

$$\Xi_{ij} = \left(1 - \xi_0 (\nu_i^{(x)} - \xi_x)^2 (\nu_j^{(y)} - \xi_y)^2 \right)^2, \quad \xi_x = 108.0 \text{ MHz}, \quad \xi_y = 102.3 \text{ MHz}, \quad (37)$$

which falls to 0.80 in the corners of the full scan range but is common to all twelve spots over the 0.37 MHz span used here [18]. With the parameters measured on this device family, $\alpha \approx 0.92$ per axis and $P_s \approx 1.70 \text{ W}$ resp. 1.62 W near 100 MHz [18]:

	C	P_{avg}	η_{axis}
x axis, 3 tones, centred, forced	2.449	0.66 W	0.636
y axis, 4 tones, $b = 98^\circ$	2.196	0.83 W	0.746
y axis, ramp, worst in the family	2.827	0.50 W	0.538
hypothetical $C = \sqrt{2}$	1.414	1.99 W	0.905
$\eta_{\text{profile}} = \eta_x \eta_y$ at the working point			47.4 %
at $C = \sqrt{2}$ on both axes, a factor 1.71			80.9 %
laser power for 10 μW in the profile			21.1 μW
for the 0.87 μW of a π pulse			1.8 μW

Assumptions: common Bragg matching for all tones, the 0.37 MHz span costing 0.007 % against the 36 MHz bandwidth by the detuned two wave solution [16]; first order only; a linear transducer, the instantaneous strain carrying the crest factor as well, so that a high C also brings on acoustic nonlinearity; isotropic diffraction, whereas the device is anisotropic and birefringent, the two agreeing qualitatively [20]; and a two tone estimate for the inter-modulation, which is optimistic once many tones contribute to the same bin.

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